Stick-together calculations

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find a common final velocity using total momentum.

SUCCESS CRITERIA - I can...

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “inelastic” and “initial” to explain your model or result. Sentence starters are optional.

inelastic: a collision that transforms some kinetic energy to other forms

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

01 Warm-up: bring back an earlier idea

Fragments of masses 3 and 5 kg separate from rest. The 3 kg piece moves at +10 m/s.
Find the other velocity and total kinetic energy.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. v_f is the shared final velocity.

Worked reasoning: A 2 kg cart at +3 m/s sticks to a 1 kg cart at rest. Total momentum 6 kg·m/s divided by total mass 3 kg gives final velocity +2 m/s.

Your turn: Mark the conserved quantity on a before-and-after sketch.
Explain why masses are added in the final denominator.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two carts stick.
Identify which masses belong in the final momentum expression and why.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Before contact: add velocity arrows.



Marzano check - circle: 1 2 3

04 Apply the physics

A 2 kg cart at +5 m/s sticks to a 3 kg cart at -1 m/s.
Find final velocity and kinetic energy transformed.
Organize the known quantities on the before-and-after model.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student averages the two starting velocities without considering mass. When would that happen to work?

Sentence starter: I notice ... . This supports ... because ... .

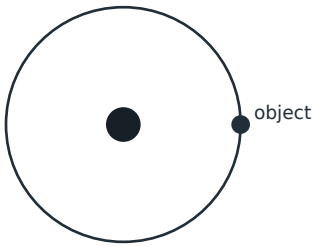
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed mass and speed, circle radius becomes 4 times as large.
Compare acceleration and net force using equations.
Label velocity and inward acceleration on the circular model.

Word help: net force = the vector sum of all forces on one object

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 0.20 kg projectile at 30 m/s sticks to a 1.80 kg cart at rest.
Find common speed and fraction of initial kinetic energy remaining. Organize the known quantities on the before-and-after model.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 1 kg cart at +6 m/s sticks to a 2 kg cart at -3 m/s.
Find final velocity. If the second velocity were -1 m/s instead, find the new result. Organize the known quantities on the before-and-after model.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3